

Topic 5. NEW DEVELOPMENTS IN MANUFACTURING.

Industry 4.0 is more than just a flashy catchphrase. A confluence of trends and technologies promises to reshape the way things are made.

(Cornelius Baur and Dominik Wee, McKinsey & Company)

DISCUSSION/SPEAKING PRACTICE

3D Printing/Additive Manufacturing vs. Traditional Manufacturing

Please study the following links:

Overview

- <https://3dprinting.com/what-is-3d-printing/>

Go to **ted.com** and watch 1 or 2 talks on 3D printing

This might be of interest as well:

https://www.ted.com/talks/taneka_jones_how_to_3d_print_human_tissue

<https://www.statista.com> (What is the value of the additive manufacturing market worldwide?)

Methods of 3D printing:

<http://3dprintingfromscratch.com/common/types-of-3d-printers-or-3d-printing-technologies-overview/>

<https://formlabs.com/blog/what-is-selective-laser-sintering/>

<https://futurism.com/engineers-have-found-a-way-to-3d-print-super-strong-aluminum/>

<https://www.sculpteo.com/blog/2016/05/25/when-is-3d-printing-the-best-solution-for-production/>

<https://www.ge.com/additive/additive-manufacturing/information/fused-deposition-modeling-technology>

Videos:

https://www.youtube.com/watch?v=9E5MfBAV_tA (Selective Laser Sintering (SLS)

Technology, 2min 30 sec)

Book:

https://assets.openstax.org/oscms-prodcms/media/documents/Additive_Manufacturing_Essentials_-_WEB.pdf

You are encouraged to do your own research in order to be well prepared for the discussion. Have you learned something about 3d printing in your lectures, projects? Is there any article, video, etc. you find especially interesting or/an informative? Please share!

Take time to explore the following key terms — understanding and using them will be essential for the upcoming discussion.

3D Printing / Additive Manufacturing Vocabulary

Term	Definition
3D Printing	A subset of additive manufacturing typically using plastic or resin materials to build parts layer by layer.
Additive Manufacturing (AM)	The process of creating objects by adding material layer by layer based on a digital model.
Binder Jetting	A process that uses a liquid binding agent to join powder particles together in a layer-by-layer fashion.
Bridging	Printing horizontal structures between two points without support underneath.
Build Platform/Print Bed	The physical platform of the 3D printer where the object is constructed. It is usually made of glass, aluminum, or other rigid materials.
Build Volume	The maximum physical size of an object that a 3D printer can produce.
CAD (Computer-Aided Design)	Software used to design 3D models before printing.
Clearance (das Spaltmaß)	Intentional space or gap between two components to ensure proper function, movement, or assembly.
Curing	The hardening of resin material using UV light after the printing process.
Delamination	A failure mode where printed layers separate or do not adhere properly.
Design for Additive Manufacturing (DfAM)	Design principles tailored to leverage the capabilities and constraints of 3D printing.
Digital Light Processing (DLP)	A printing technique similar to SLA, but using a digital light projector to cure resin.
Direct Metal Laser Sintering (DMLS)/SLM (Selective Laser Melting)	A type of metal 3D printing that uses a laser to sinter metal powder into solid objects.
Dual Extrusion	A printer feature that allows two different filaments or colors to be printed simultaneously.
Electron Beam Melting (EBM)	A metal printing process using an electron beam to fully melt metal powder.
Extruder	The part of a 3D printer that pushes material through a heated nozzle.
Filament	A spool of thermoplastic material used as feedstock in FDM 3D printers.
Functional Prototype	A working model created through 3D printing for testing design and usability.
Fused Deposition Modeling (FDM)	A common 3D printing method where thermoplastic material is extruded layer by layer.
Fused Filament Fabrication (FFF)	A thermoplastic filament is heated, melted and deposited layer-by-layer. Is the most common and affordable 3D

	printing technology today, slowly replacing binder jetting.
G-code	The language used to control CNC machines and 3D printers, instructing movement and operation.
Heatbed	A heated build platform that helps prevent warping by keeping the base layers warm.
Infill	The internal structure of a printed object, typically a lattice pattern for strength and weight savings.
Layer Thickness	The height of each individual layer in a 3D printed object, affecting resolution and speed.
Material Jetting	A 3D printing method where droplets of material are deposited layer by layer to build parts.
Mesh	A network of polygons representing the surface geometry of a 3D object.
Nozzle	The tip of the extruder through which molten filament is deposited.
Open-source Printer	A 3D printer with publicly available design and software, allowing customization and modification.
Overhang	A part of the printed model that extends outward and requires support to print correctly.
Post-processing	Finishing operations such as sanding, curing, or painting applied after printing.
Print Speed	The rate at which the printer head moves while printing, affecting quality and time.
Rapid Prototyping	Quickly creating scale models or parts for testing and evaluation using 3D printing.
Resin	A liquid polymer that solidifies when exposed to UV light, used in SLA and DLP printing.
Resolution	The level of detail in a printed object, influenced by layer height and nozzle diameter.
Retraction	The process of pulling back filament to prevent oozing when the print head moves.
Selective Laser Sintering (SLS)	A printing process where a laser fuses powdered material into solid structures.
Shell	The outer perimeter or wall of a 3D printed object.
Slicing	The process of converting a 3D model into layers and generating the corresponding G-code instructions.
Stereolithography (SLA)	A 3D printing technique using a laser to cure liquid resin into hardened plastic.
STL File	A file format that represents 3D geometry and is commonly used in 3D printing.
Stringing	Thin threads of material left between printed parts due to oozing during travel moves.
Support Structures	Temporary structures added during printing to support overhangs or bridges.
Tolerance	The permissible limit of variation in a physical dimension, critical in precision printing. Compare to Clearance!!
Topology Optimization	A design process using algorithms to reduce material use while maintaining strength.
Warping	A defect in 3D printing where edges lift or curl due to uneven cooling.

X-Y Axis	The horizontal plane in which the print head or build platform moves.
Z-axis	The vertical movement axis of a 3D printer, controlling layer height.



In groups of 3 or 4 have a 15-20-minute discussion addressing all agenda points mentioned in the setting. A representative of your group is going to summarize your arguments and conclusions and present them afterwards.

WATCH THE FOLLOWING VIDEO

MILL TALK: What Is Industry 4.0 and How Did We Get Here? With MIT Professor David Hardt <https://www.youtube.com/watch?v=ttfMEXGdh1s> (1 hour 16 minutes long; the questions are based on the first 21 minutes).

Answer the following questions:

1. What is industry 1.0?

2. What is industry 2.0?

3. What is industry 3.0?

4. What is industry 4.0?

5. What is the difference between digitization and digitalization?

6. When and how was the term "Industry 4.0" coined?

7. How do we keep the employees current?

8. Industry 4.0 is a combination of which technologies?

9. Which countries can be considered pioneers in the field of industry 4.0? Why?

10. Describe the graph at time stamp 20.33-20.43. What is Moore's law?

TOPIC RELATED VOCABULARY

Task 1. What do these acronyms stand for? Give a short description/definition of the notion.

1. CAD –

2. CAE –

3. CAM –

4. DFM –

5. DFR –

6. C2C –

7. CIM –

8. CNC –

9. ANN –

10. CAPP –

**Task 2. Match the phrases describing new trends in manufacturing with their definitions.
Can you provide examples/expand the definitions?**

1. Concurrent engineering	A a methodology that might appear logical at first, in practice often leads to waste
2. Virtual prototyping	B a methodology where the design and manufacture of products are integrated with a view toward optimizing all elements involved in the life cycle of the products
3. Six sigma	C a software-based method that uses advanced graphics and virtual-reality environments to allow designers to view and examine a part in detail
4. Agile manufacturing	D a set of techniques and tools for process improvement; a process in which 99.99966% of all opportunities to produce some feature of a part are statistically expected to be free of defects
5. Smart manufacturing	E employs computer-integrated manufacturing, high levels of adaptability and rapid design changes, digital information technology, and more flexible technical workforce training
6. Reshoring	F green design and manufacturing considers all possible adverse environmental impacts of materials, processes, operations, and products, so that they can all be taken into account at the earliest stages of design and production
7. Environmentally conscious design and manufacturing	G a production method aimed primarily at reducing times within the production system as well as response times from suppliers and to customers
8. Lean production	H a manufacturing methodology that places an extremely strong focus on rapid response to the customer – turning speed into a key competitive advantage
9. Sequential engineering	I the practice of transferring a business operation that was moved overseas back to the country from which it was originally relocated

MATERIALS TESTING

Read the text and complete the tasks. There is a glossary at the end of the text.

1. Design and engineering would be almost impossible without detailed knowledge of the materials used. The more sophisticated a design, the more effective a machine or construction, the greater the role of the right steel, plastic or concrete. The properties of these materials or components are determined through materials testing. The various test methods are divided into two major groups: they can either be destructive or non-destructive. Destructive methods are usually conducted on specifically shaped test pieces to evaluate the basic properties of a given material. In some cases, prototypes of components are also subject to destructive tests to define the working envelope of the part under certain load conditions. Non-destructive tests are often used in maintenance to check the health of a component and determine if it has to be replaced. But even after the breakdown of a part, a non-destructive investigation can often help to find the reason for the failure.

DESTRUCTIVE TESTING

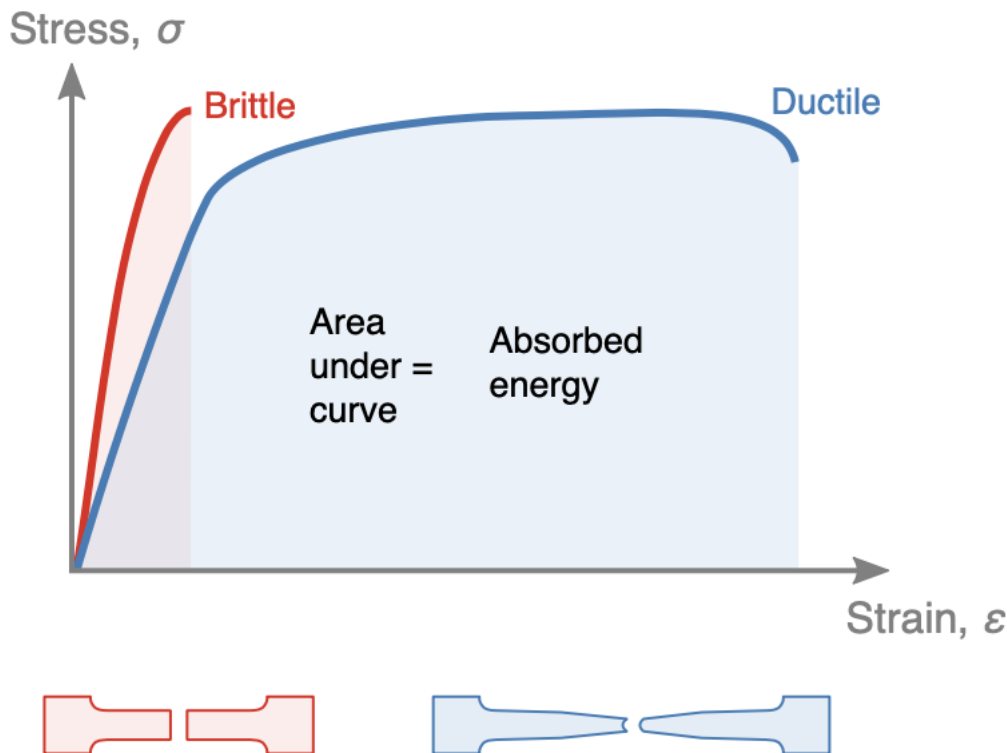
2. The most common destructive tests are tension, compression and bending tests. Since all three involve a linear motion, they can be run on a single, so-called universal testing machine. The machine consists of a strong frame and two either hydraulically or electromechanically operated pistons with grips for the test piece. Sensors record the applied force and the resulting deformation.

3. For a tensile test, a standard test piece, a flat bar or a rod with enlarged ends for the grips, is mounted in the test machine and pulled apart at a constant rate. The applied force is measured by a load cell and the extension by an extensometer. Since both measurements depend on the size of the test piece, the load is normalised by calculating the stress, i.e. force per cross section, and the extension by converting it into strain or the change in length in relation to the original length.

4. The resulting stress versus strain or tensile curve has a characteristic shape, and changes in the curve relate to changes in the material's structure. For metals, the first part of the curve is steep and linear showing the elastic deformation of the test piece. If the load were to be released, the part would return to its original shape. The Young's module of the material can be calculated from the gradient of this section. With rising strain the curve reaches a point where it 'breaks' or flattens. This so-called yield point marks the beginning of a plastic or permanent deformation. Further down the curve the stress will continue to rise with the strain until it suddenly starts dropping. Now the material begins to neck, i.e. thin down locally until it finally breaks.

5. Modulus of elasticity, yield strength, ultimate tensile strength, ultimate elongation, ductility and many other properties relevant for engineering can be derived from a series of tensile tests. Compression and bending tests also produce distinct curves revealing additional properties. Depending on the material, for example, metals, ceramics or plastics, certain parts of these curves might be pronounced or missing entirely. Since the failure of a machine or structure often means the actual breaking of a certain component, the characteristics of a material during fracture are of great importance. The Charpy impact test is designed to analyse these characteristics. At the base of the machine, a small bar with a notch in the middle is held in an anvil. A pendulum swings down and breaks the bar on

impact. The energy absorbed by the impact can be calculated by comparing the start and end deflection of the pendulum. Tougher materials will absorb more energy. Toughness varies largely with temperature and the transition from a brittle state requiring very little energy to break to a tough, ductile state can be very sharp.



https://commons.wikimedia.org/wiki/File:Stress_strain_comparison_brittle_ductile.svg

The following video illustrates the tensile test:

<https://www.youtube.com/watch?v=D8U4G5kcpcM> (by Prof. Dr.-Ing. Rainer Schwab, HS Karlsruhe), 9 min.

This 13-minute video offers more explanation on the stress vs strain curve:

<https://www.khanacademy.org/science/in-in-class11th-physics/in-in-mechanical-properties-of-solids/in-in-stress-strain-and-modulus-of-elasticity/v/stress-vs-strain-curve>

6. While a single load case such as an impact or a constant force might not result in an immediate failure, several repeated stress cycles can very well lead to a breakage. Such a fatigue fracture is caused by small cracks. The local stress is concentrated at the tip of these cracks. When they rise over the yield strength, the repeated stress cycles force the crack to widen and grow until the part breaks. Among several other methods, the fatigue resistance of a material can be determined with a rotating cantilever beam test where a test rod is bent and rotated at the same time. Fatigue testing is very complex and depends on the desired parameters, for example, the number of cycles for a given stress, the maximum stress for a given number of cycles or the maximum stress for which no failure occurs.

7. Most of the tests mentioned above will result in the destruction of the test specimen. One of the least damaging destructive tests is the hardness test. In a simple scratch test, the probe is compared to a range of standard materials with increasing hardness. More widespread are indentation tests, where a specially shaped tip is pressed on the material with a certain force. The hardness is then determined from the surface area or depth of the indentation. common tests are the Brinell test using a hardened steel ball, the Vicker's test with a pyramid-shaped diamond and the Rockwell test with a diamond cone or a steel sphere.

NON-DESTRUCTIVE TESTING

8. The variety of methods used for non-destructive testing is even wider. Most of these tests are designed to detect discontinuities in the material or component like small cracks, cavities, porosity or wall thinning from corrosion to name but a few. Non-destructive testing allows the tester to assess the state of a component without affecting its form or function. It is therefore often used in maintenance, quality control or security checks. With non-destructive tests it is possible to analyse not only mechanical properties but also the material's microstructure, its composition, magnetic or electrical properties.

9. Common test methods can be divided into six major categories: visual tests determine, for example, cracks, reflectivity or surface finish. A large category involves tests which use penetrating radiation to find out about an object's composition or to search for inclusion or voids. Electromagnetic and electric methods are used for similar purposes, but also for ascertaining hardness or state of cure. Sonic and ultrasonic test devices can also look inside a component and are relatively easy to handle and inexpensive. Thermal and infrared examination searches for thermal conductivity, plating thickness or composition while chemical and analytical tests can detect surface anomalies, grain size or help to identify alloys.

10. Both destructive and non-destructive materials testing are becoming more and more important. When engineering is taken to its limits in search for the most elegant, most cost-effective solution, the precise knowledge of a material's properties is crucial.

Matthias Meier, INCH 24, #1 2020, pp. 24-25.

GLOSSARY

alloy	Legierung	load cell	Kraftaufnehmer, Kraftmessdose
anvil	Amboss	maintenance	Wartung
ascertain	bestimmen, ermitteln	neck	Hier: einschnüren
brittle	spröde	notch	Kerbe, Nut
cavity	Lunker, Hohlraum	piston	Kolben
Charpy impact test	Kerbschlagbiegeversuch	property	Eigenschaft
compression	Druck, Stauchung	rotating cantilever beam test	Umlaufbiegeversuch
concrete	Beton	specimen	Muster, Probe
conduct	Ausführen, durchführen	strain	Dehnung
conductivity	Leitfähigkeit	stress	Spannung
cone	Kegel	tensile	Zug-
cure	Aushärtung	tension	Zug(-spannung)
deflection	Auslenkung, Ablenkung	toughness	Zähigkeit
ductility	Dehnungsvermögen	ultimate elongation	Bruchdehnung
extensometer	Dehnungsmessgerät	ultimate tensile strength	Bruchfestigkeit
fatigue fracture	Ermüdungsbruch, Dauerbruch	void	Pore, Fehlstelle
gradient	Steigung	working envelope	Arbeitsbereich, Einsatzbereich
grip	Spannzeug, Probenhalter	yield point	Streck-, Dehngrenze
indentation	Abdruck, Einkerbung	yield strength	Streckfestigkeit, Umformfestigkeit
load case	Lastfall	Young's module	Elastizitätsmodul

a) Find words in the text with the same or similar meanings as the following.

1. operational range (paragraph 1) –
2. the application of “pulling” forces so as to stretch a material (paragraph 2) –
3. the application of “pushing” forces to different points on a material so as to reduce its size in one or more directions (paragraph 2) –
4. the behavior of a slender element subjected to an external load applied perpendicularly to a longitudinal axis of the element (paragraph 2) –
5. the moving component that is contained by a cylinder (paragraph 2) –
6. test in which a sample is subjected to a controlled tension until failure (paragraph 3) –
7. a sample of a material used for testing (paragraph 7) –
8. the measure of void (“empty”) spaces in a material (paragraph 8) –
9. a thin coating of gold, silver, or other metal (paragraph 9) –
10. a mixture of chemical elements of which at least one is a metal (paragraph 9) –

b) Answer the following questions.

1. When are destructive and non-destructive tests usually conducted?
2. Describe the Charpy impact test.

The following video might help you:

<https://www.youtube.com/watch?v=tpGhqQvftAo> (5 min)

3. Which indentation tests do you know?
4. What are different types of non-destructive tests?
5. Do you know of any other destructive and non-destructive tests?

Properties of Materials

Materials and substances can be identified by listing their physical and chemical properties. Physical properties include for example colour, odour and taste. Hydrogen for instance is a colourless, odourless and tasteless gas. Chemical properties include the stability towards heat as well as reactive behaviour in the presence of other substances, for instance the solubility of one substance in another.

Exercise 1¹

Read the text on aluminium properties and underline the words that describe or are related to properties.

Aluminium is a relatively soft, durable, lightweight, ductile and malleable metal with appearance ranging from silvery to dull gray, depending on the surface roughness. It is nonmagnetic and does not easily ignite. A fresh film of aluminium serves as a good reflector (approximately 92%) of visible light and an excellent reflector (as much as 98%) of medium and far infrared radiation. Aluminium is a good thermal and electrical conductor. Its melting point is 660.32 °C and its boiling point is 2519 °C.

Exercise 2 Fill in the gaps

solvent – poisonous – retain – fluids – properties – smell – gaseous – compound – solid – pressure – oxygen

1. Oxygen is an odorless gas, in other words, one cannot _____ it.
2. Air is an example of matter in its _____ state.
3. Most elements are in the _____ state at 25 °C and at one atmosphere _____.
4. Water is a _____ made up of the elements hydrogen and _____.
5. In a compound, iron does not always _____ its magnetic _____.
6. Since chlorine is a _____ gas, precautions must be taken not to breathe it.
7. Liquids and gases are also called _____, because they can conform to the shape of their container.
8. A usually liquid substance capable of dissolving or dispersing one or more other substances is called a _____.

¹ Source reference: Adapted from <http://en.wikipedia.org/wiki/Aluminium> on 30 August 2013

The Properties of Solids

If an object is *light*, it does not weigh very much; if it is *heavy*, it weighs a great deal. These terms are, of course, relative. A uranium atom is *quite* heavy compared with a hydrogen atom, but it's light compared with a snowflake. Thus, we can say

A uranium atom	weighs <i>more</i> THAN	a hydrogen atom.
A uranium atom	weighs <i>less</i> THAN	a snowflake.
A uranium atom is	<i>relatively</i> heavy.	
A helium atom is	heavier THAN	a hydrogen atom.
A helium atom is	not AS heavy AS	an oxygen atom.
A snowflake is	<i>quite</i> heavy.	
An elephant is	<i>extremely</i> heavy.	
Coarse sandpaper is	rougher THAN	fine sandpaper.
	MORE abrasive THAN	

A Systematic Overview of Adjectives Used to Describe Properties

light	heavy	<i>How much does it weigh?</i>
brittle	tough	<i>How easily will it break?</i>
soft	hard	<i>Can it be dented or scratched easily?</i>
flexible, pliable	rigid	<i>How easily will it bend?</i>
rough	smooth	<i>How easily will the surface produce friction when rubbed?</i>
soluble	insoluble	<i>Will it dissolve easily?</i>
combustible flammable inflammable	non-combustible non-flammable non-inflammable	<i>How easily will it catch fire and burn?</i>
elastic	plastic	<i>Will it return to its original shape when the deforming force is removed?</i>

A word which is difficult to translate into German.

Flimsy is a word which is difficult to translate into German. Depending on the context, the term flimsy means "hauchdünn, zart, leicht, schwach, dürrig, nicht haltbar, von schlechter Qualität, durchsichtig, schwach, fadenscheinig." Paper tissues (like Kleenex™ or Tempo™) are much flimsier than paper towels; they are thinner and fall apart more quickly when wet. This doesn't mean they are useless, although in some situations the word flimsy can have a negative connotation: badly made using inferior materials.

Can something pass through it?

water-tight, water-proof, water-resistant	If something is water-tight or water-proof, water cannot pass through it or into it; however, if it is water-resistant, water can pass through or into it, although with difficulty.
porous	Porous describes a material full of pores (minute holes) through which pure liquids, solutions or gases can pass, e.g. a brick is porous; a piece of glass is non-porous.
perforated	Perforated describes a flat piece of material that has many small holes (perforations) which have been made deliberately (or accidentally) to pass right through, e.g. an ear-drum can be perforated.
permeable	Permeable materials are capable of allowing fluid substances to pass through, usually by a process of diffusion, e.g. sand is more permeable to water than clay (it allows water to pass through more easily). A permeable membrane allows water and solutes to pass through it. Semi-permeable membranes are permeable to solvents but not to solutes; semi-permeable membranes are used in osmosis. Impermeable substances do not allow fluid substance to pass through, e.g. an impermeable membrane does not allow solvents or solutes to pass through.

Can light pass through it?

Transparent materials allow light to pass through. Translucent materials allow light to pass through them only diffusely, they cannot be seen through. Opaque materials transmit very little light, and therefore reflect, scatter, or absorb most of it.

Different forms of solids

ADJECTIVES:	granular	powdery	crystalline	amorphous
NOUNS:	granules	powder	crystals	

Properties of fluids

The **viscosity** of a liquid is the resistance of a liquid to flow. This can be measured, and in Germany the *Engler degrees* scale is used. This list is in order of viscosity, with free-flowing being the least viscous and thick or viscous the most.

free-flowing thin oily creamy thick viscous

Some substances are between the solid and liquid forms: **gel (jelly) cream paste**

Gases are also **fluids**. They can be **reactive** or **inert**.

Properties of surfaces

shiny	↔	matt	smooth	↔	rough	abrasive	pitted
bright	↔	dull	coarse	↔	fine	corrugated	

You should be sure that you understand the following words, as they often cause difficulty:

Malleability: It is easy to roll a malleable material into a new shape. A malleable material does not fracture easily under pressure. Gold is extremely malleable. It is possible to roll gold into very thin sheets. Copper is very malleable and so is lead. Glass is not all malleable and nor is cast iron. It is very easy to fracture glass with a hammer. Cast iron also fractures easily.

Ductility: It is easy to draw a ductile material. It does not fracture and it retains its new shape. Copper is extremely ductile. Tin is very ductile and so is aluminum. Steel is not very ductile and nor is lead. It is very difficult to draw lead into thin wire because it fractures easily.

Elasticity: An elastic material stretches easily under stress. However, remove the stress and it does not retain its new shape. It regains its original shape. Rubber is extremely elastic. Some steels are quite elastic. Glass is not at all elastic and nor is cast iron.

Durability: A durable material does not corrode easily. Under normal conditions, glass is very durable and so are plastics. Among the metals, chromium is extremely durable and so is platinum. Gold is quite durable and so is aluminum. Cast iron is not very durable and nor is steel.

Exercise 3 Complete the descriptions below with words learned in this unit:

1. Glass is a _____ solid which usually has a _____ surface.
2. Coarse sandpaper has a very _____ or _____ surface.
3. Some cardboard is _____ to give it extra strength.
4. The inside of an old camera has a _____ surface.
5. A golf ball has a _____ surface.
6. Copper gets a rather _____ surface if it is not polished.
7. At room temperature, mercury is a _____, silvery substance.
8. If you plan to drive through Death Valley in the summer, you should use a _____ oil in your car. But if you are driving in Alaska in the winter, you will need a _____ oil.
9. Silicon and germanium are good _____.
10. Helium, neon, argon, krypton, xenon, and radon are _____ gases.
11. Hydrogen is a very _____ gas; in the presence of oxygen, it is easily ignited in an explosive way.
12. Salt is highly _____ in warm water.
13. When a rubber band gets old, it loses its elasticity and becomes _____.
14. Since cotton is a _____ material which absorbs moisture, it is suitable for summer clothing.
15. Glass has a(n) _____ structure, but sugar has a(n) _____ structure.

Exercise 4 Find a word that fits the following definitions:

1. Lasts a long time despite frequent use; will not be worn away easily by chemical action.
2. The tendency to fracture without much deformation.
3. Something that is easily broken or damaged or destroyed.
4. Thin and easily broken or damaged, poorly made.
5. Describes an object which does not allow the passage of light of wavelength in the visible range.
6. The property of a material that allows it to be used for the removal of matter by scratching and grinding.

7. Something which does not bend and is not flexible, unyielding stiff and hard.
8. Describes a material or substance which easily catches fire and produces flame.
9. The property of a material which does not produce friction easily.

Exercise 5

1. Why are roads "salted" in the winter?
2. Why do water pipes burst in the winter?
3. Why does food cook faster in salty water?
4. If the metal lid (*Deckel*) on a jar (*Dose*) is stuck, hold it under a hot water tap. Why?
5. If a ping-pong ball is squashed, put it in warm water. Why?
6. Why do guitars often have to be tuned on stage? (Note: Guitars have "strings.")
7. Why does pouring boiling water into a glass often make the glass shatter?
8. Why does an aerosol can explode when thrown into a fire?
9. Why do icebergs float?

Exercise 6. Complete the table of opposites.

↔	
light	
brittle	
soft	
thick	
flexible, pliable	
rough, abrasive	
soluble	
inflammable	

↔	
elastic	
opaque	
shiny	
a <u>dull</u> red	
free-flowing	
a good <u>conductor</u>	
<u>coarse</u> sandpaper	
a <u>reactive</u> gas	

Exercise 7 Fill in the gaps.

- Gold is a good _____, since electricity flows through it easily, and ceramic is a good _____. Silicon is one of the best-known _____.
- If light passes through a material, but you cannot see through it, it is called _____.
- If a material resists wear, it is _____ ("verschleißfest").
- If you see the word _____ on a box, you should not treat it roughly. The object(s) inside could break easily.
- Caution! Don't put sharp objects into your ear. You could _____ your ear-drum.
- _____ electricity can cause your hair to stand on end.
- After many years of use, cars driven in the desert get a _____ surface.
- Wire is made of a _____ material like copper or gold.
- Rubbing alcohol is a volatile fluid which _____ quickly.